Title Page

SBT-DF204 - Reconstructing Chrome Web History

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Date of Submission: 12/10/2026

Introduction

Digital forensic investigations involve examining electronic records to understand activities that may have taken place on a computer or other digital device. Web browser history is one source of evidence because it can reveal websites visited, the time of access, navigation patterns, and records of downloaded files.

This investigation focused on a Google Chrome History database obtained from a device associated with a suspected drug trafficking case. The examination was carried out using Kali Linux and SQLite to identify relevant browsing activities, recover timestamps, and reconstruct the sequence of events recorded in the database.

Particular attention was given to Craigslist posting activity, Gmail communications, cryptocurrency-related websites, and a downloaded image named proof_of_payment.png. The purpose was to determine how these activities might relate to one another while ensuring that conclusions remained within the limits of the available digital evidence.

Objectives

The main objective of this investigation was to examine the supplied Chrome History database and reconstruct an accurate, evidence-based account of the recorded browser activity.

The specific objectives were to:

Preserve the original Chrome History database and verify the integrity of its forensic working copy using SHA-256 hashing.

Identify and examine the relevant SQLite database tables and their fields.

Extract browser-history records and convert Chrome timestamps into Coordinated Universal Time (UTC).

Examine navigation transitions and relationships between recorded webpage visits.

Investigate browsing records associated with Craigslist posting activity and Gmail communications.

Identify cryptocurrency-related browsing and examine the available download evidence.

Develop a chronological timeline of significant events and assess the possible relationships between them.

Evaluate the strength and limitations of the evidence before reaching a forensic conclusion.

Laboratory / Tools Used

The investigation was conducted in a Kali Linux virtual machine running through VMware Workstation. The Kali terminal provided the environment for organising evidence, executing forensic commands, and examining the Chrome History database.

SQLite3 was used to examine the database structure and extract records through SQL queries. Commands such as file, stat, cp, chmod, and sha256sum were used during evidence preparation, identification, preservation, and integrity verification.

The examination mainly involved the urls, visits, downloads, and downloads_url_chains tables. SQL queries were used to connect related records, interpret navigation information, convert timestamps, and export the browser-history results into a CSV file for further examination.

Executive Conclusion

The forensic examination of the supplied Chrome History database identified a sequence of browsing activities involving Craigslist, Gmail, Imgur, Blockchain.com, Kraken, and Mempool.space. These records were recovered from a database containing 57 URL entries, 74 individual browser visits, and one recorded file download.

The investigation established that the browser accessed Craigslist pages associated with creating and managing an advertisement titled "Cheaper than Rx Supplements." Subsequent records showed Gmail pages containing related subject lines, followed by an image download named proof_of_payment.png and visits to cryptocurrency transaction and exchange platforms.

Several records were connected through matching posting identifiers, cryptocurrency transaction references, and related URLs. These relationships support the possibility that the activities formed part of a connected sequence involving an advertisement, communications, and payment-related browsing.

However, the evidence does not independently establish that controlled drugs were sold, that cryptocurrency payment was completed, or that a particular individual carried out the recorded activities. The findings therefore support the reconstruction of a potentially relevant browsing sequence rather than definitive proof of the suspected offence.

Investigation Methodology

Evidence Acquisition and Preservation
The investigation began with the acquisition of the supplied Chrome History database. A dedicated forensic investigation directory was created in Kali Linux to organise the original evidence, working copy, screenshots, SQL results, and report materials.

The original database was preserved in the Original_Evidence folder, while a separate copy was placed in the Working_Copy folder for examination. This approach allowed the investigation to proceed without unnecessary changes to the preserved evidence.

SHA-256 hashes were calculated for both database files. The results were identical, and each file had a size of 196,608 bytes. This confirmed that the working copy matched the preserved original at the time of verification.

Database Identification and Examination

The working database was examined using SQLite3. Initial checks identified the file as an SQLite 3.x database, while the SQLite integrity check returned ok, indicating that no structural errors were detected.

The database structure was then examined through SQL queries. Seventeen tables were identified, with particular attention given to those containing website addresses, individual browsing visits, downloaded files, and download URL chains.

The relevant fields were examined before extracting records. This helped establish how the stored information could be connected and interpreted during the investigation.

Browser History Extraction and Timestamp Conversion

SQL queries were used to connect records from the visits and urls tables through their corresponding identifiers. This made it possible to retrieve visit IDs, URL IDs, website titles, timestamps, transition values, and related browsing information.

Chrome timestamps were converted from WebKit time into UTC using the following calculation:

```
datetime(
  v.visit_time / 1000000 - 11644473600,
  'unixepoch'
)
```

The converted timestamps were used to arrange the recorded browser visits in chronological order.

The examination established that the 74 recorded visits occurred on 19 April 2022, between 13:54:18 and 15:02:56 UTC.

Navigation and Transition Examination

The navigation analysis focused on the transition and from_visit fields to determine how the browser moved between recorded webpages.

The basic transition type was obtained using:

```
v.transition & 255
```

This revealed navigation types associated with links, generated searches, and form submissions.

The from_visit field was also examined to identify relationships between successive browsing events. These findings were particularly useful when reconstructing the Craigslist posting process and analysing navigation involving Gmail and cryptocurrency-related websites.

Download Examination and Evidence Correlation

The downloads and downloads_url_chains tables were examined to identify downloaded files and their recorded sources.

One download record was identified for an image named proof_of_payment.png. Its timestamps, recorded file size, completion status, and associated Imgur URL were examined.

The download information was then compared with relevant browser visits. Similar comparisons were made between Craigslist posting identifiers, Gmail page titles, and cryptocurrency transaction references.

Timeline Reconstruction and Documentation

After examining the relevant browser records, a complete timeline was exported into a CSV file.

The exported file contained 74 visit records and one header row. A separate SHA-256 hash was generated for the exported file to provide an integrity reference.

The resulting timeline allowed the significant activities to be reviewed in chronological order while separating recorded facts from their possible interpretations.

Investigation Findings

Finding 1: Chrome History Database Integrity

The investigation confirmed that the supplied Chrome History file was a valid SQLite database. Both the preserved original and forensic working copy were 196,608 bytes, with matching SHA-256 hash values.

The database integrity check also returned a successful result, allowing the examination to continue without any identified structural errors.

The database contained 57 URL records, 74 visits, one download entry, and one associated download URL-chain record.

Evidence References: E01, E02, E03, and E08.

Finding 2: Craigslist Posting Activity

The earliest significant browsing activity involved searches for Craigslist, followed by navigation to Craigslist's Baltimore pages.

Between 13:55:10 and 13:58:22 UTC, several records showed the browser progressing through pages for selecting a posting category, creating an advertisement, adding location information, previewing the posting, and reaching a confirmation page.

Visit ID 20 recorded access to a posting management page, while Visit ID 21 contained a listing titled "Cheaper than Rx Supplements." Both records referenced the Craigslist posting identifier 7473121658.

The matching identifier provides a meaningful connection between the posting management page and the recorded listing. However, the browser history does not reveal the complete advertisement or establish that controlled substances were being sold.

Evidence References: E09, E10, E11, and E16.

Finding 3: Gmail Communication-Related Activity

Following the Craigslist activity, the browser recorded navigation to Gmail and Google account pages.

At 14:04:12 UTC, Visit ID 43 showed a Gmailpage titled "cheaper than Rx supplements - order." Subsequent visits contained Gmail URLs with compose-related parameters.

Later, at 14:56:47 UTC, another Gmail record included a subject line containing a cryptocurrency transaction identifier.

The relationship between the Craigslist advertisement title and the Gmail subject lines suggests that the browser accessed communications potentially associated with the earlier posting.

However, the available records do not reveal the complete email messages, identify all participants, or confirm that any specific message was successfully sent.

Evidence References: E11, E12, E13, and E17.

Finding 4: Imgur Image and Download Activity

At 14:56:52 UTC, the browser accessed an Imgur page associated with the image identifier dTgrkP7.

The database also contained a download record for proof_of_payment.png, which began at 14:56:58 UTC and ended at 14:57:10 UTC.

The download was recorded as completed, with 33,844 bytes received. Its URL-chain information linked the file to an Imgur-hosted PNG image.

These details establish a relationship between the Imgur browsing activity and the recorded image download.

Although the filename suggests that the image may have been intended to represent payment evidence, the downloaded image itself was not examined. Its contents and authenticity therefore remain unverified.

Evidence References: E12, E14, and E15.

Finding 5: Cryptocurrency Transaction-Related Browsing

The browser recorded searches for a blockchain explorer shortly after the image download.

At 14:59:06 UTC, the history showed navigation to a transaction-related page on Blockchain.com. An additional record showed access to a cryptocurrency address page.

The transaction identifier found in the Blockchain.com records also appeared in the Gmail-related records and later in the Mempool.space URL.

This repeated identifier provides a direct link between the recorded email-related browsing and the cryptocurrency transaction lookups.

Nevertheless, the browser history alone cannot establish who controlled the cryptocurrency address, whether funds were transferred, or whether any payment was associated with the suspected advertisement.

Evidence References: E12, E13, E17, and E18.

Finding 6: Kraken Exchange Activity

Following the blockchain-related browsing, the history recorded Google searches for Kraken and visits to the platform's sign-in and device-approval pages.

Additional records showed access to trading, funding, and transaction-history interfaces.

At 15:01:33 UTC, the browser also recorded navigation through Kraken-related redirect pages to Mempool.space, where the previously identified cryptocurrency transaction reference appeared again.

These records establish that the browser accessed several cryptocurrency-related services during the period under investigation.

However, they do not independently establish successful account authentication, ownership of an exchange account, or completion of a financial transaction.

Evidence References: E13, E17, and E18.

RECONSTRUCTED UTC TIMELINE

The following timeline presents the significant events recovered during the investigation. All timestamps refer to 19 April 2022 (UTC).

Stage	UTC Timestamp	Visit / URL ID	Artefact	Observed Fact	Interpretation and Limit
Initial browsing	13:54:18	V1 / U1	Google Search	Search for Craigslist recorded.	Establishes the beginning of the relevant browsing sequence.
Posting preparation	13:55:05	V11 / U7	Craigslist	Browser reached a posting-type selection page.	Indicates entry into a posting workflow.
Category selection	13:55:10	V14 / U10	Craigslist	Category selection page recorded with a form-submission transition.	Supports interaction with a posting form.
Posting creation	13:55:15	V15 / U11	Craigslist	Browser accessed the create-posting page.	Suggests preparation of an advertisement.
Location details	13:58:13	V16 / U12	Craigslist	Map-related posting page recorded.	Indicates progression through the posting process.
Posting confirmation	13:58:22	V19 / U8	Craigslist	Posting confirmation page recorded.	Supports completion of a posting workflow, but not independently its public availability.
Posting management	13:58:27	V20 / U15	Craigslist	Posting management page containing identifier 7473121658.	Connects the browser to a specific posting record.

Stage	UTC Timestamp	Visit / URL ID	Artefact	Observed Fact	Interpretation and Limit
Advertisement page	13:58:31	V21 / U16	Craigslist	Listing titled "Cheaper than Rx Supplements" recorded.	Matches the identifier found in the management page.
Email access	13:59:31	V42 / U33	Gmail	Gmail inbox page recorded.	Establishes email-related browsing.
Email thread	14:04:12	V43 / U34	Gmail	Page titled "cheaper than Rx supplements - order."	Suggests a relationship with the earlier listing.
Compose-related navigation	14:04:35	V44 / U35	Gmail	Gmail URL contained a compose parameter.	Indicates compose-related navigation, not proof of a sent message.
Transaction reference	14:56:47	V48 / U37	Gmail	Email-page title contained a cryptocurrency transaction identifier.	Provides a reference for comparison with later transaction lookups.
Image-hosting activity	14:56:52	V50 / U39	Imgur	Browser accessed an Imgur image page.	Provides context for the subsequent recorded download.
Image download	14:56:58	Download ID 1	Chrome Downloads	Download of proof_of_payment.png began.	Filename suggests possible payment-related material.
Download completion	14:57:10	Download ID 1	Chrome Downloads	Download completion recorded; 33,844 bytes received.	Establishes the recorded completion of the image download.
Transaction lookup	14:59:06	V56 / U43	Blockchain.com	Cryptocurrency transaction page recorded.	Transaction reference matches other browsing artefacts; payment remains unverified.
Address examination	14:59:21	V58 / U44	Blockchain.com	Cryptocurrency address page recorded.	Indicates address-related browsing without proving ownership.

Stage	UTC Timestamp	Visit / URL ID	Artefact	Observed Fact	Interpretation and Limit
Exchange access	14:59:47	V61 / U46	Kraken	Sign-in page recorded.	Establishes an attempt to access the exchange interface, not successful authentication.
Device approval	15:00:21	V63 / U47	Kraken	Device-approval page recorded.	Provides additional account-access context.
Funding interface	15:01:21	V66 / U50	Kraken	Funding overview page recorded.	Shows navigation to a funding-related interface.
Transaction exploration	15:01:33	V70 / U54	Mempool.space	Transaction page containing the previously recorded identifier.	Links the transaction reference across browsing records.
Transaction history	15:02:05	V71 / U55	Kraken	History/ledger page recorded.	Shows access to a transaction-history interface, not proof of a specific transfer.
Return to Gmail	15:02:56	V74 / U37	Gmail	Browser returned to the transaction-related email page.	Establishes further email-related browsing after the cryptocurrency lookups.

Lab Folder Structure and Evidence Preparation

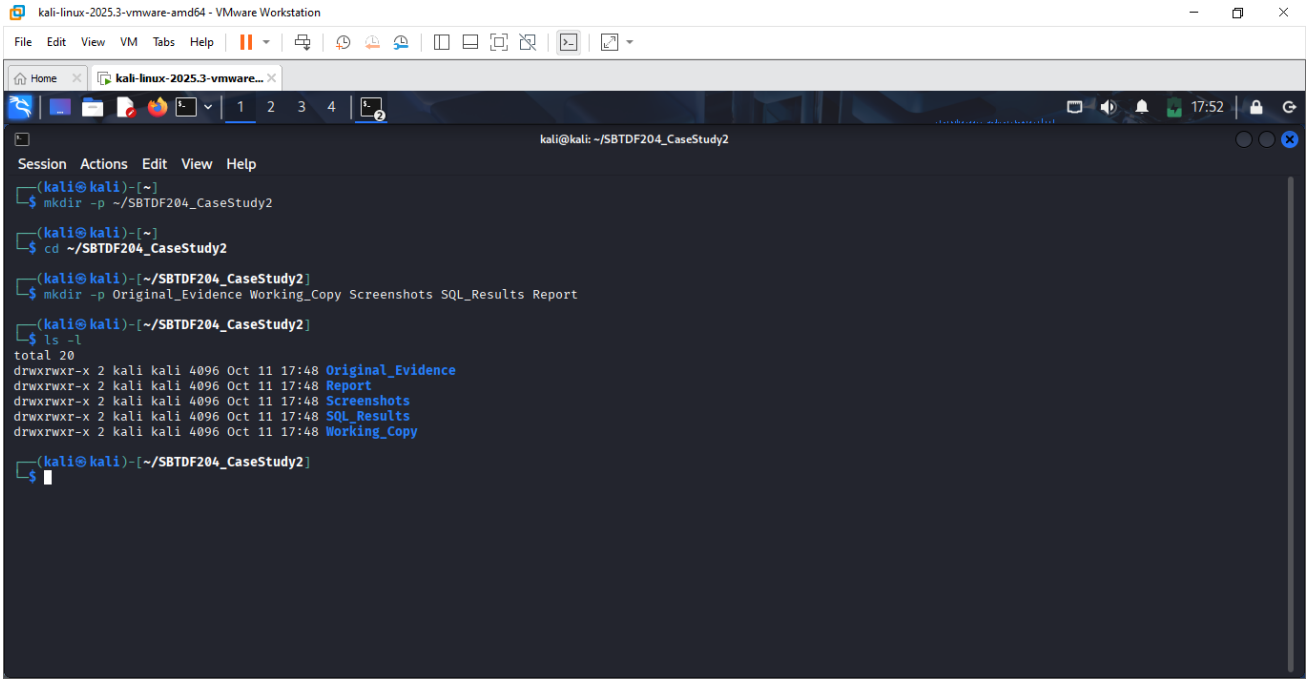


Fig 1.0 Forensic Investigation Directory Structure.

Explanation

The screenshot shows the preparation of the investigation workspace in Kali Linux, with separate folders created for the original evidence, working copy, screenshots, SQL results, and report files. This arrangement helped keep the investigation materials organised and reduced the possibility of confusing the preserved evidence with files created during analysis.

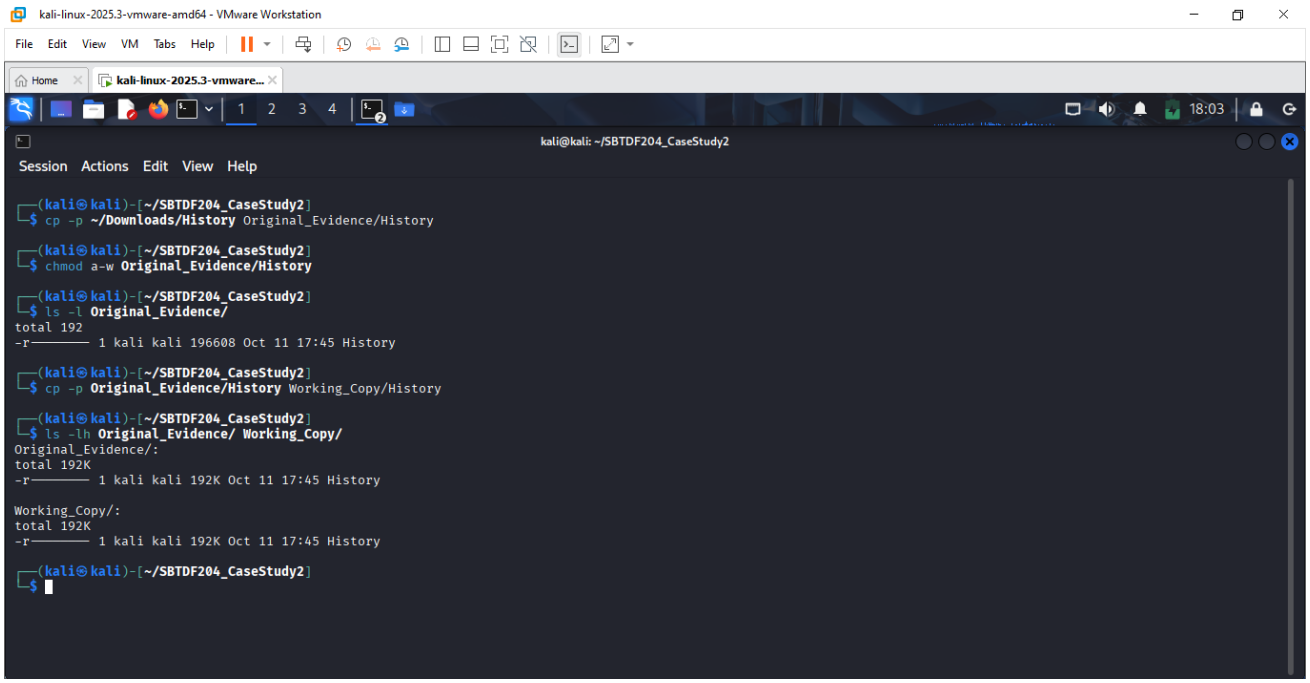


Fig 1.1 Original Chrome History Evidence Preservation and working copy.

Explanation

The screenshot documents the preservation of the Chrome History database in a dedicated evidence folder and the creation of a separate working copy. The original evidence was made read-only to protect it from unintended modification, allowing the database examination to be performed using the copied file.

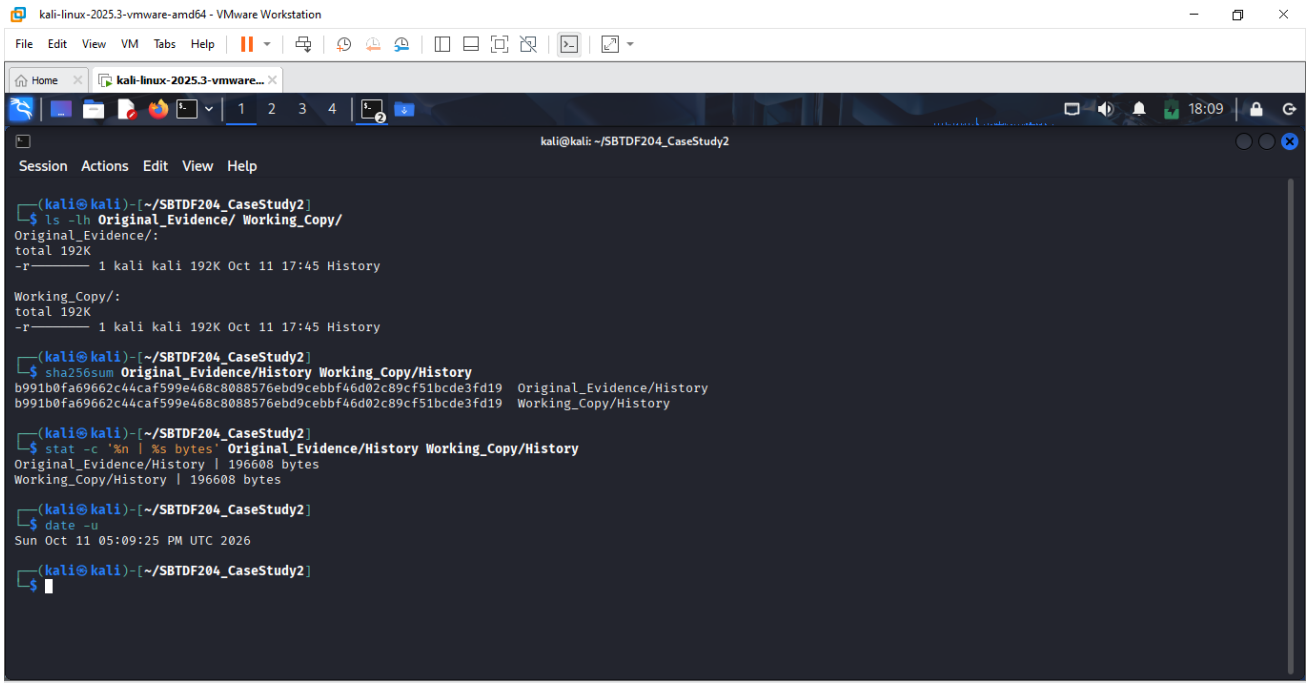


Fig 1.2 Chrome History SHA-256 Integrity Verification.

Evidence Log E01 — Evidence Acquisition and Integrity Verification

Field	Evidence Details
Evidence ID	E01
Source Record	Original_Evidence/History and Working_Copy/History
Finding	The original Chrome History database and its forensic working copy were verified to have identical SHA-256 hash values. Both files were 196,608 bytes, confirming that the copied evidence matches the preserved **original.**
Why It Matters	Matching SHA-256 hashes demonstrate that the working copy is identical to the preserved evidence file. This supports evidence integrity and ensures that the investigation can proceed without modifying the original database.
SQL or Screenshot Reference	Fig 1.2 — Chrome History SHA-256 Integrity Verification

Explanation

The screenshot shows the SHA-256 verification performed on the original Chrome History database and its working copy. Both files were confirmed to have identical hash values and a size of 196,608 bytes, demonstrating that the working copy matched the preserved evidence at the time of verification.

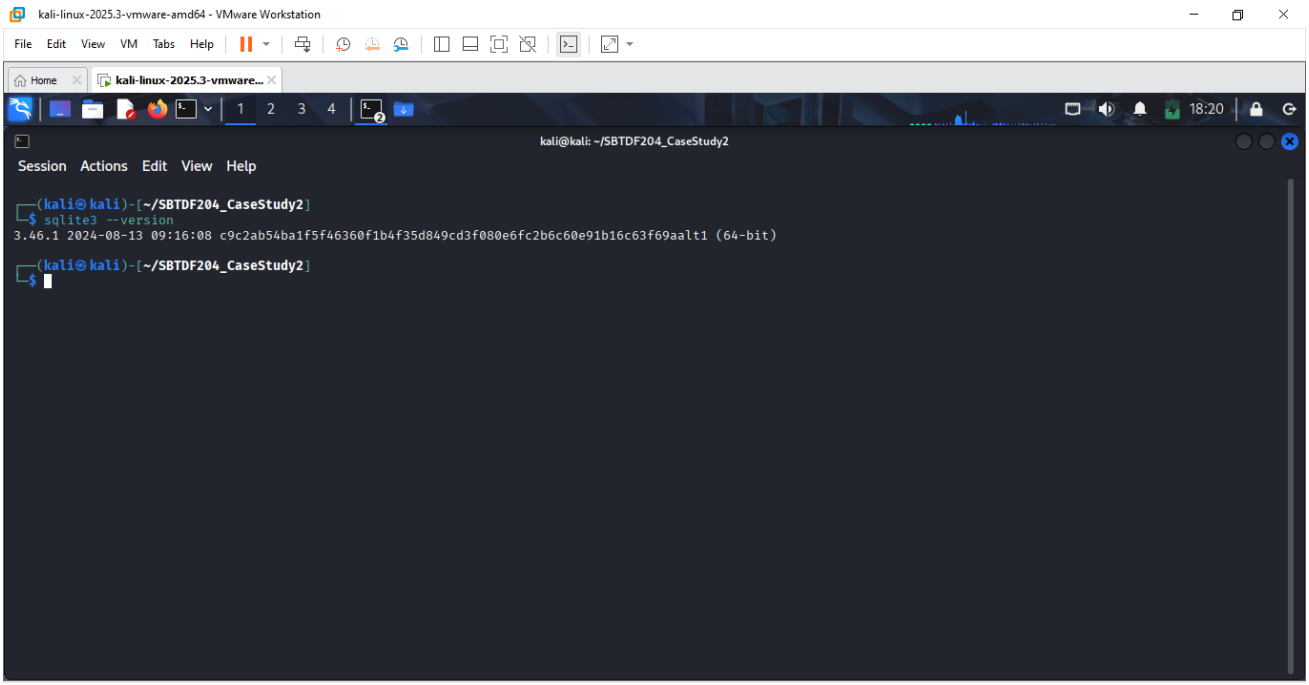


Fig 1.3 Verify SQLite Installation.

Explanation

The screenshot shows the Kali Linux terminal being used to check the availability of SQLite3 before examining the Chrome History database. The successful version output confirmed that SQLite3 was installed and available for executing the SQL queries required during the investigation.

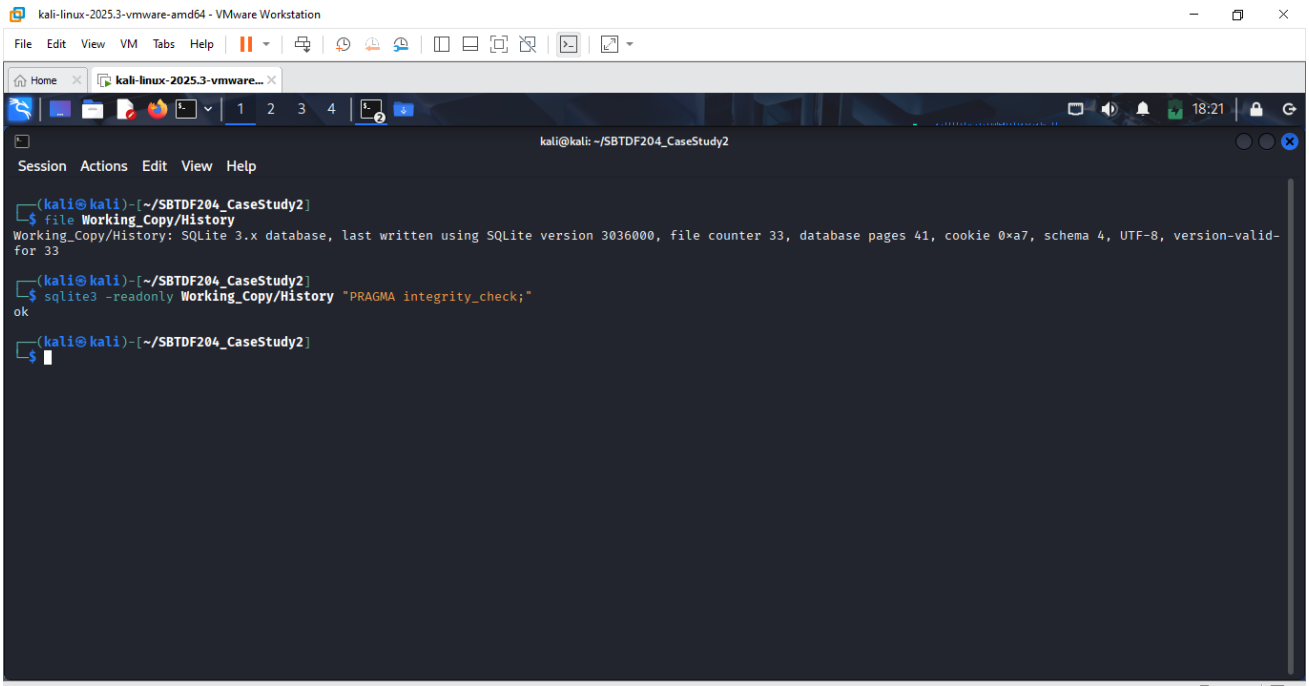


Fig 1.4 Chrome History Database Identification and Integrity Check.

Explanation

The screenshot presents the examination of the working History file using the file command and SQLite's

integrity-check function. The file was identified as an SQLite 3.x database, while the integrity check returned ok, indicating that no structural errors were detected.

Evidence log — E02

Field	Evidence Details
Evidence ID	E02
Source Record	Working_Copy/History
Finding	The file was identified as an SQLite 3.x database. The PRAGMA integrity_check command returned ok.
Why It Matters	Confirms that the working database is recognised as an SQLite database and that the integrity check detected no structural errors. This supports the reliability of subsequent database examination.
SQL or Screenshot Reference	Fig 1.4 — Chrome History Database Identification and Integrity Check

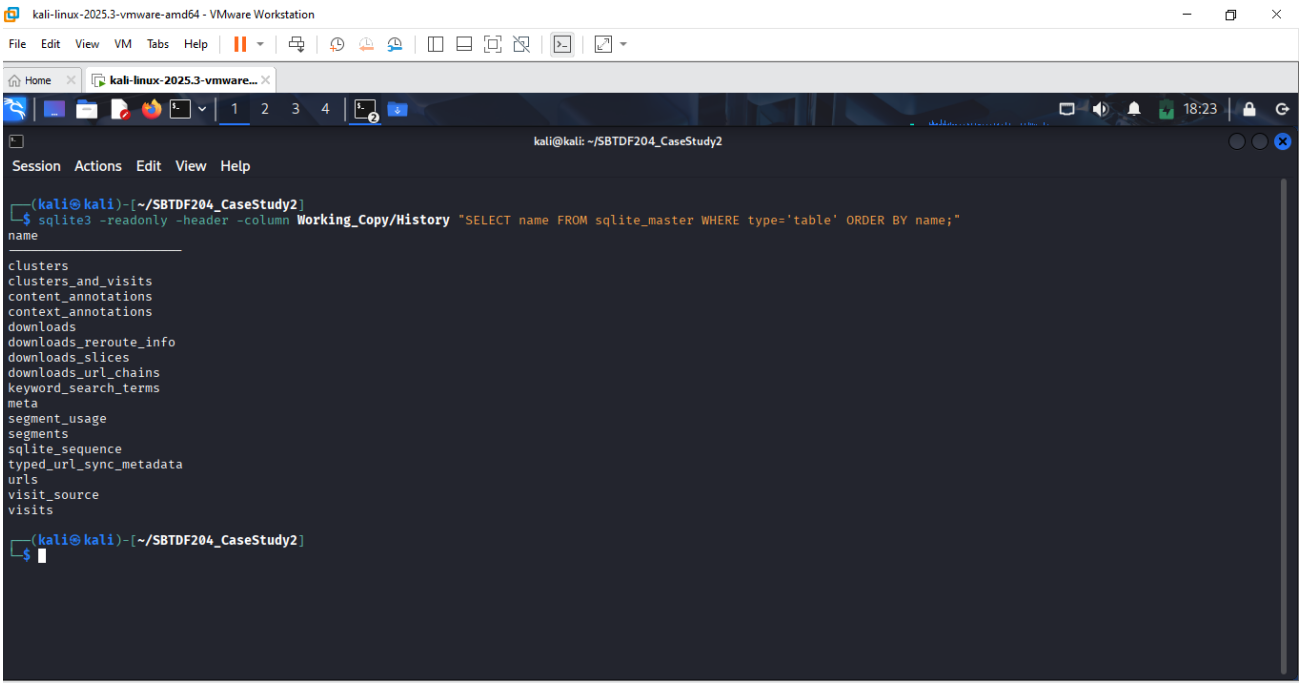


Fig 1.5 Chrome History SQLite Database Tables.

Evidence Log — E03

Field	**Evidence Details**
Evidence ID	E03
Source Record	sqlite_master table within Working_Copy/History
Finding	The database contains 17 tables, including urls, visits, downloads, downloads_url_chains, and keyword_search_terms.

Why It Matters	Identifying the available tables establishes the database structure and highlights the records needed to reconstruct browser activity, navigation sequences, searches, and downloads.
SQL or Screenshot Reference	Fig 1.5 — Chrome History SQLite Database Tables

Explanation

The screenshot shows an SQL query used to identify the tables stored within the Chrome History database. Seventeen tables were returned, including urls, visits, downloads, and downloads_url_chains, which were selected for further examination because of their relevance to browsing activity and downloaded files.

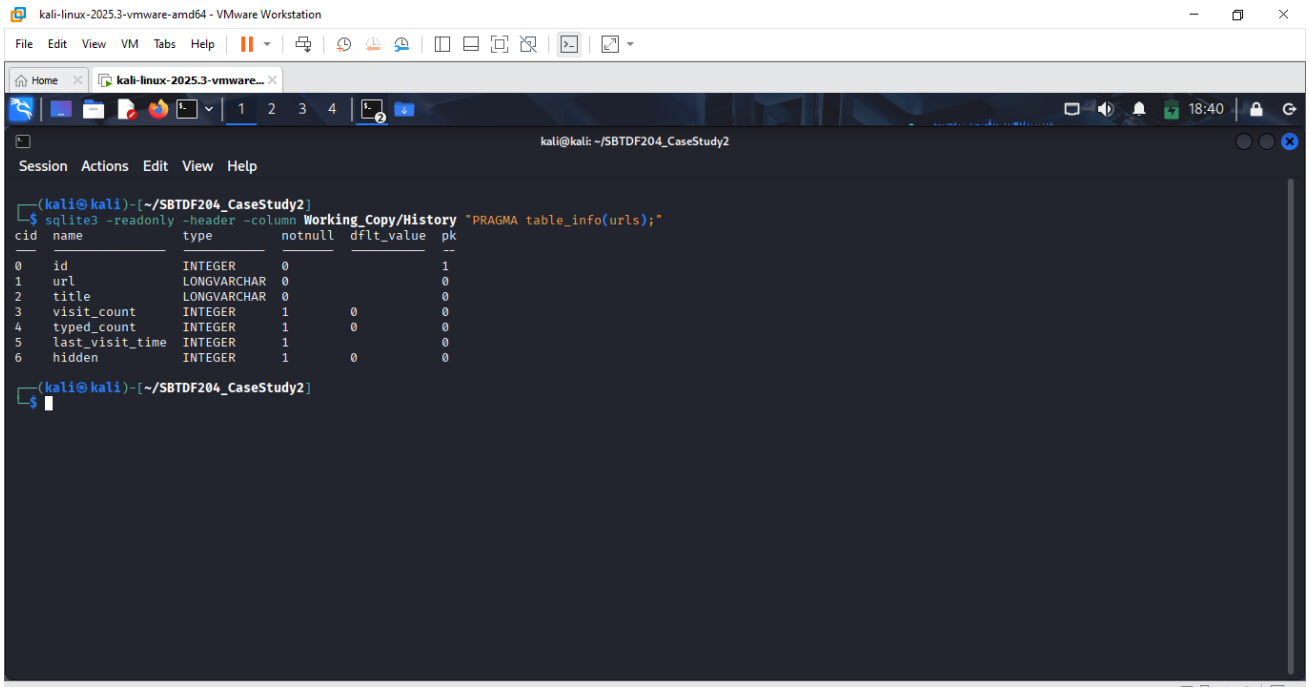


Fig 1.6 Chrome History URLs Table Structure.

Evidence Log — E04

Field	Evidence Details
Evidence ID	E04
Source Record	urls table in Working_Copy/History
Finding	The SQL query identified seven columns: id, url, title, visit_count, typed_count, last_visit_time, and hidden. These fields describe recorded website addresses, page titles, visit counts, and related browser-history information.
Why It Matters	The identified fields provide the structure needed to examine websites recorded in Chrome's history. They will help identify relevant URLs and page titles associated with the suspected activity and connect them to individual visit records.

SQL or Screenshot Reference	Fig 1.6 — Chrome History URLs Table Structure

Explanation
The screenshot displays the structure of the urls table, revealing seven fields used to store website addresses, page titles, visit counts, and related information. These fields provided the foundation for identifying recorded websites and connecting them with individual browser visits during the investigation.

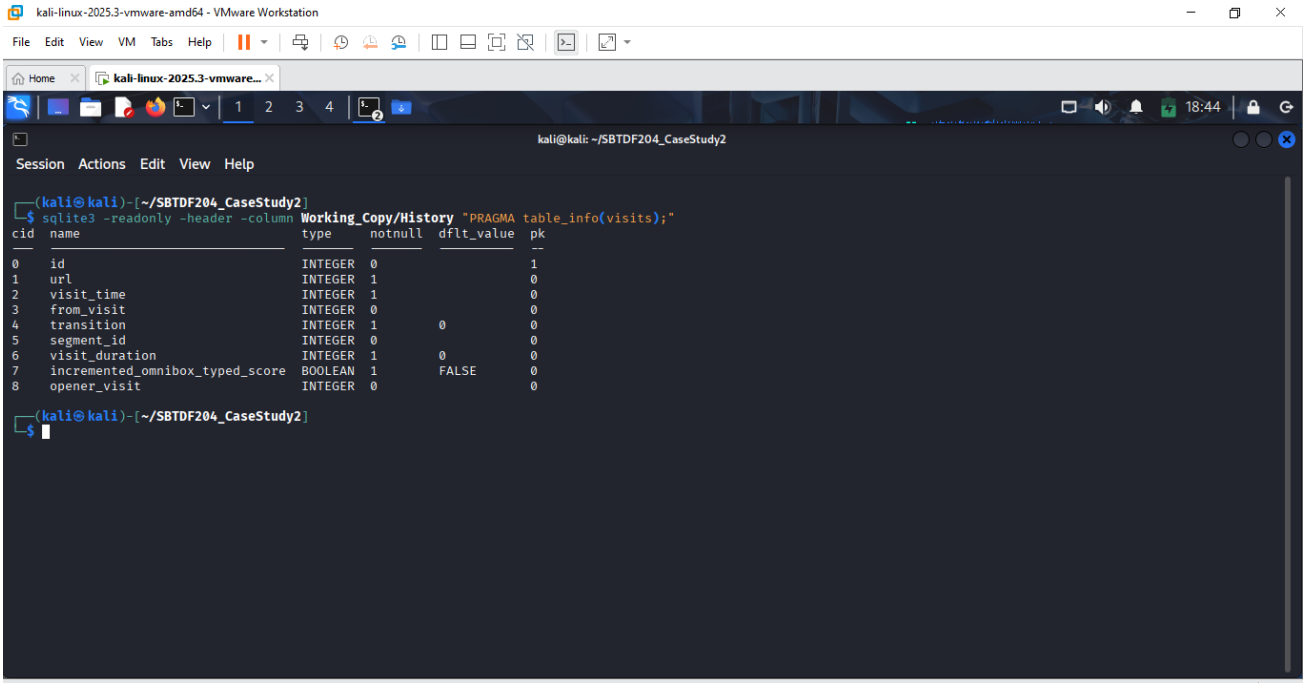


Fig 1.7 Chrome History Visits Table Structure.

Evidence Log — E05

Field	Evidence Details
Evidence ID	E05
Source Record	visits table in Working_Copy/History
Finding	The SQL query identified nine columns: id, url, visit_time, from_visit, transition, segment_id, visit_duration, incremented_omnibox_typed_score, and opener_visit. These fields contain the structure needed to examine individual browser visits and navigation relationships.
Why It Matters	The visits table is essential for reconstructing browser activity chronologically. Its timestamps, navigation transitions, and linked visit identifiers will help establish the sequence of recorded events and distinguish different navigation methods.
SQL or Screenshot Reference	Fig 1.7 — Chrome History Visits Table Structure

Explanation
The screenshot shows the nine fields identified within the visits table, including visit identifiers, timestamps,

navigation transitions, and references to preceding visits. Examining these fields helped establish how individual browsing events could be arranged chronologically and linked to related webpages.

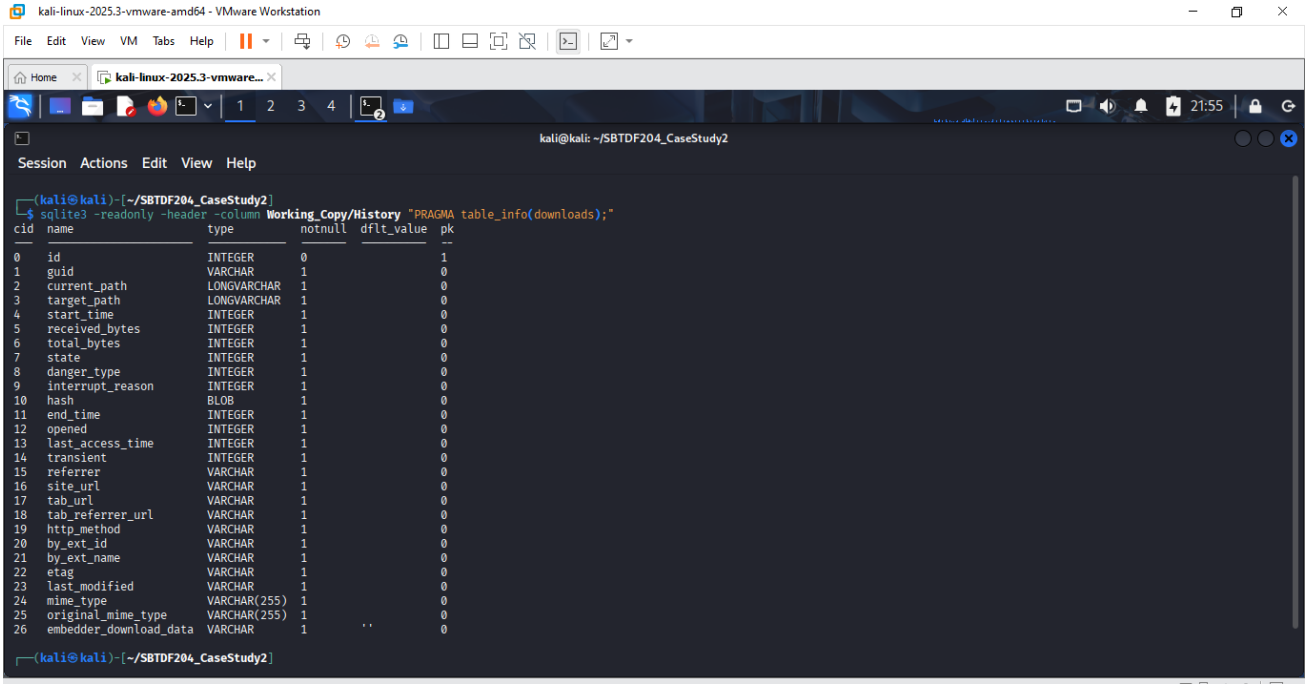


Fig 1.8 Chrome History Downloads Table Structure.

Evidence Log — E06

Field	Evidence Details
Evidence ID	E06
Source Record	downloads table in Working_Copy/History
Finding	The SQL query identified 27 columns, including id, guid, current_path, target_path, start_time, received_bytes, total_bytes, state, end_time, opened, referrer, and mime_type. These fields provide information about downloaded files, their locations, timestamps, sizes, and download status.
Why It Matters	The identified fields will help investigators examine download records that may relate to the suspected activity. They can provide information about when files were downloaded, where they were saved, and whether the database records indicate successful or interrupted downloads.
SQL or Screenshot Reference	Fig 1.8 — Chrome History Downloads Table Structure

Explanation

The screenshot presents the results of an SQL query examining the downloads table, which contained 27 fields associated with file downloads. These fields included file paths, timestamps, byte counts, download status, and file information, providing the structure needed to investigate the download record identified later in the examination.

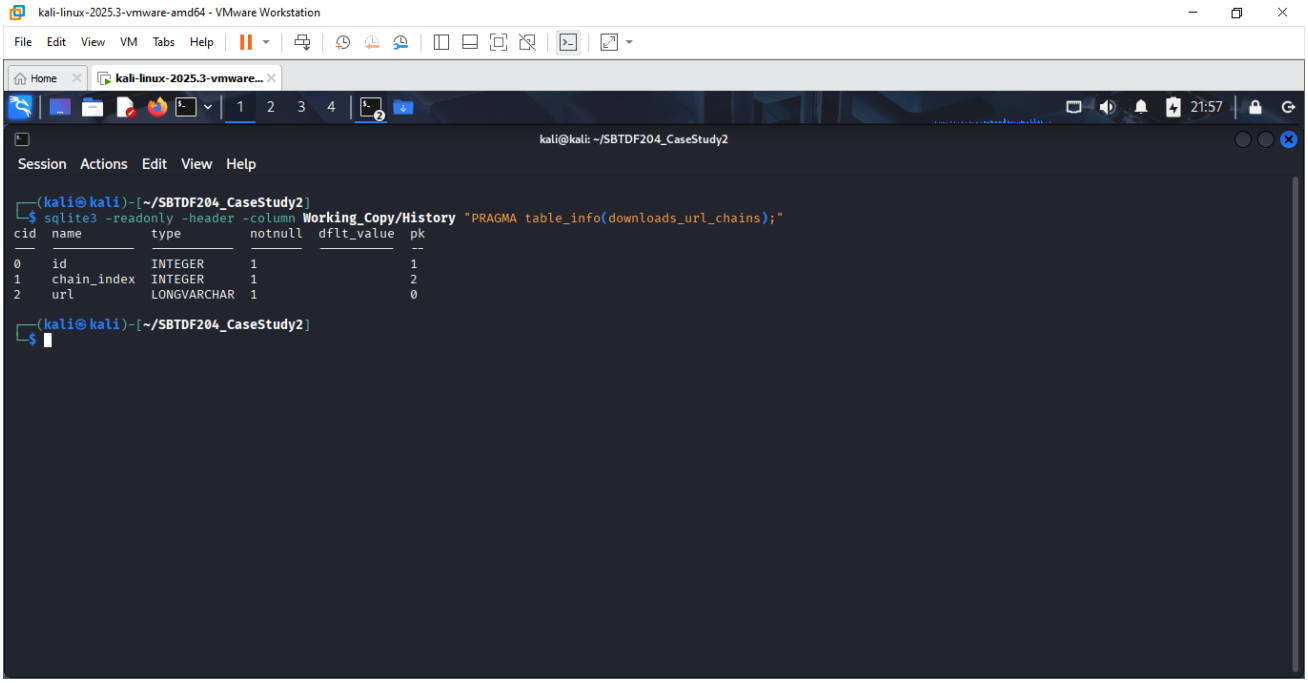


Fig 1.9 Chrome History Downloads URL Chains Table Structure.

Evidence Log — E07

Field	Evidence Details
Evidence ID	E07
Source Record	downloads_url_chains table in Working_Copy/History
Finding	The SQL query identified three columns: id, chain_index, and url. These fields provide the structure for associating download records with their corresponding URLs and the order of URLs within a download chain.
Why It Matter	This table can help reconstruct the source and navigation path associated with downloaded files. When linked to the downloads table, it may provide additional context about where a download originated and whether multiple URLs were involved.
SQL or Screenshot Reference	Fig 1.9 — Chrome History Downloads URL Chains Table Structure

Explanation

The screenshot shows the three fields identified in the downloads_url_chains table: id, chain_index, and url. These fields were important because they allowed the recorded download to be linked to its associated web address and helped establish whether additional URLs appeared in the download chain.

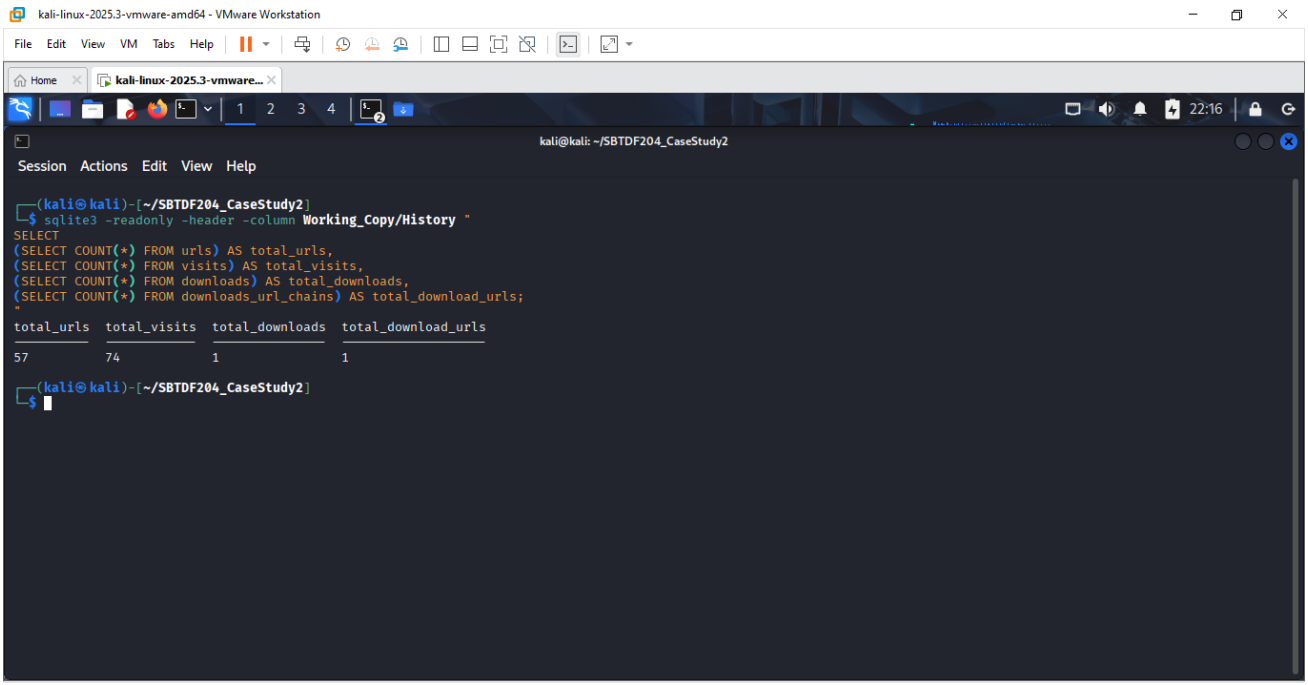


Fig 2.0 Chrome History Database Record Counts.

Evidence Log — E08

Field	Evidence Details
Evidence ID	E08
Source Record	urls, visits, downloads, and downloads_url_chains tables in Working_Copy/History
Finding	The SQL query identified 57 URL records, 74 individual browser visits, one download record, and one download URL-chain record.
Why It Matters	These results establish the quantity of available browser-history evidence. The recorded visits provide a basis for reconstructing browsing activity, while the download and URL-chain records may help identify files relevant to the investigation.
SQL or Screenshot Reference	Fig 2.0 — Chrome History Database Record Counts

Explanation

The screenshot shows an SQL query used to determine the number of records stored in the four main tables examined during the investigation. The results identified 57 URL records, 74 visits, one download, and one download URL-chain entry, establishing the quantity of available browser-history evidence.

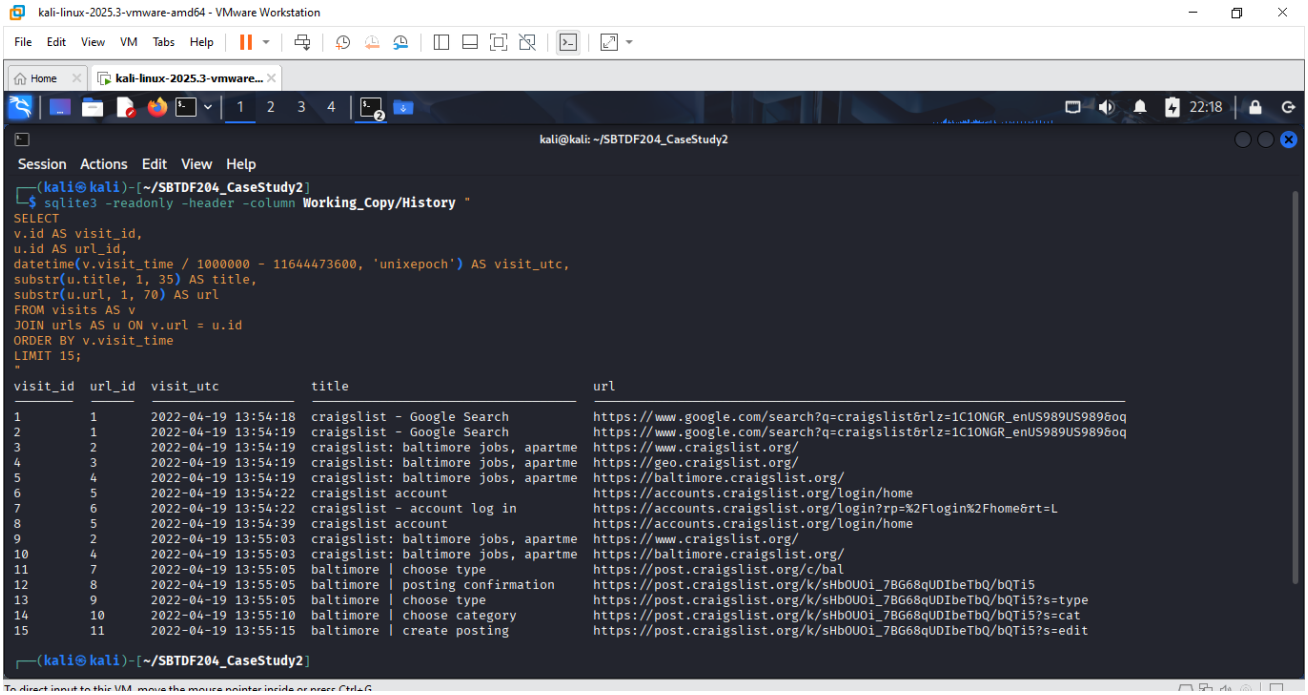


Fig 2.1 Chrome Browser Visit History and UTC Timestamp Conversion.

Evidence Log — E09

Field	Evidence Details
Evidence ID	E09
Source Record	Joined visits and urls tables; visit IDs 1–15, URL IDs 1–11
Finding	The first 15 browser visits occurred between 13:54:18 and 13:55:15 UTC on 19 April 2022. The records show Google searches for Craigslist, navigation to Craigslist and its Baltimore pages, access to account-related pages, and subsequent visits to pages associated with creating a Craigslist posting.
Why It Matters	The records establish an early sequence of browser activity that may be relevant to the suspected advertisement. The posting-related pages provide a starting point for examining whether the browser history supports a posting workflow, although they do not independently establish that an advertisement was successfully published.
SQL or Screenshot Reference	Fig 2.1 — Chrome Browser Visit History and UTC Timestamp Conversion

Explanation

The screenshot displays the first 15 browser visits extracted from the joined visits and urls tables, with their timestamps converted into UTC. The records show initial Google searches for Craigslist, navigation to Baltimore-related pages, and access to posting-related interfaces, providing the starting point for reconstructing the browsing timeline.

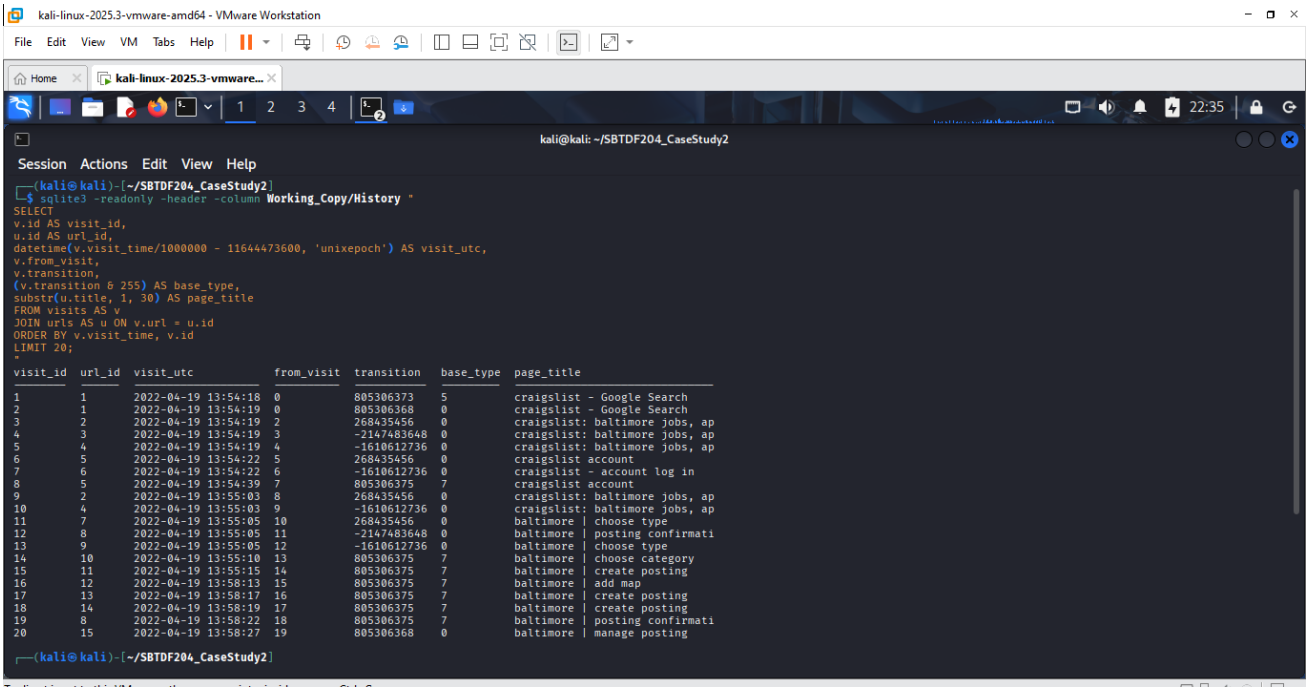


Fig 2.2 Chrome Browser Navigation and Transition Analysis.

Evidence Log — E10

Field	Evidence Details
Evidence ID	E10
Source Record	Joined visits and urls tables; Visit IDs 1–20
Finding	The SQL query identified different Chrome navigation transitions, including base types 0, 5, and 7. Visit 1 recorded generated search activity, while Visit IDs 14–19 contained transition value 805306375, corresponding to base type 7 (form submission). The from_visit values also established navigation relationships between successive Craigslist posting pages.
Why It Matters	The transition values and linked visit records provide evidence of how the browser navigated through the Craigslist website. The repeated form-submission transitions support an interactive posting workflow rather than a sequence consisting entirely of ordinary page visits.
SQL or Screenshot Reference	Fig 2.2 — Chrome Browser Navigation and Transition Analysis

Explanation

The screenshot shows an examination of the first 20 browser visits, including their transition values and links to preceding records. Several entries contained transition type 7, indicating form submissions associated with the Craigslist posting process and providing evidence of interactions beyond ordinary webpage navigation.

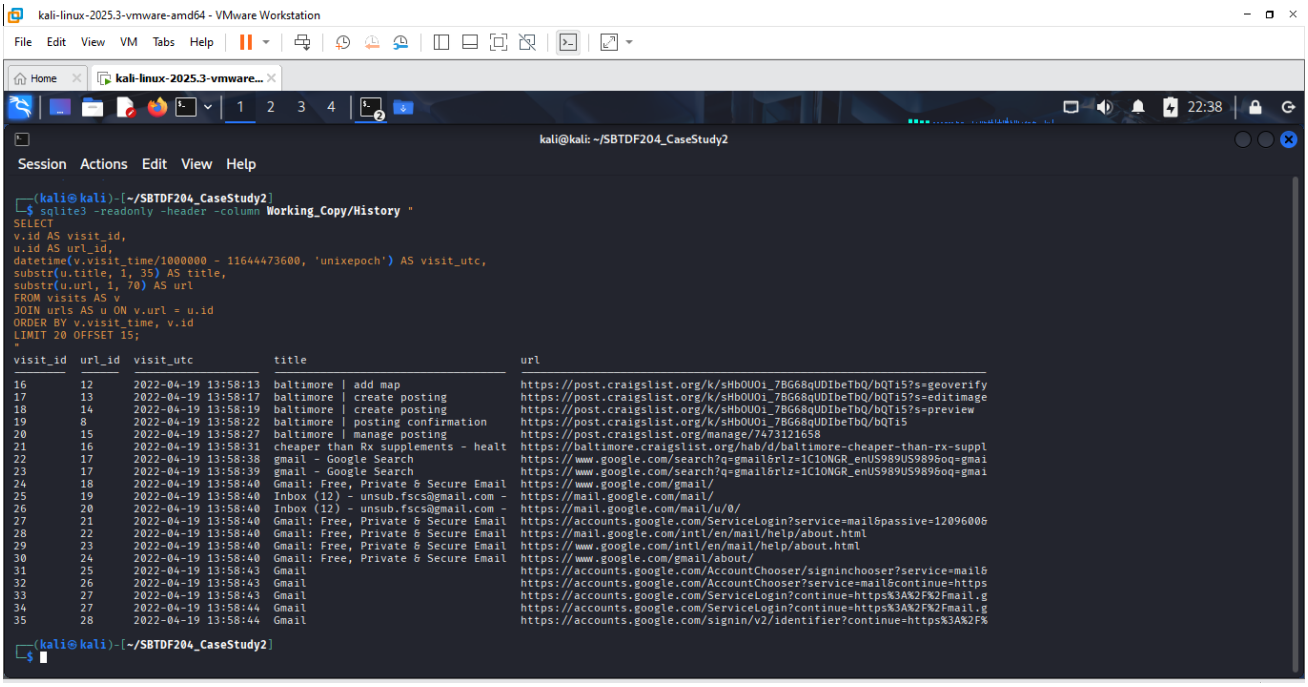


Fig 2.3 Chrome Browser History Records 16–35.

Evidence Log — E11

Field	Evidence Details
Evidence ID	E11
Source Record	Joined visits and urls tables; Visit IDs 16–35, URL IDs 8–28
Finding	Between 13:58:13 and 13:58:31 UTC on 19 April 2022, the browser recorded Craigslist pages involving map addition, posting creation, posting confirmation, management, and a listing titled “Cheaper than Rx Supplements.” From 13:58:38 to 13:58:44 UTC, additional records showed Google searches for Gmail, Gmail inbox pages, and Google account-related navigation.
Why It Matters	The sequence provides evidence of a Craigslist posting workflow followed shortly by email-related browser activity. The listing title may be relevant to the suspected advertisement, while the Gmail records establish subsequent access to email-related pages. However, the history alone does not establish the advertisement's contents or prove that any particular message was sent.
SQL or Screenshot Reference	Fig 2.3 — Chrome Browser History Records 16–35

Explanation

The screenshot presents browser visits 16–35, showing continued Craigslist activity involving posting creation, confirmation, and management. A listing titled "Cheaper than Rx Supplements" was recorded before the browser moved to Gmail and Google account pages, establishing a chronological relationship between posting and email-related browsing.

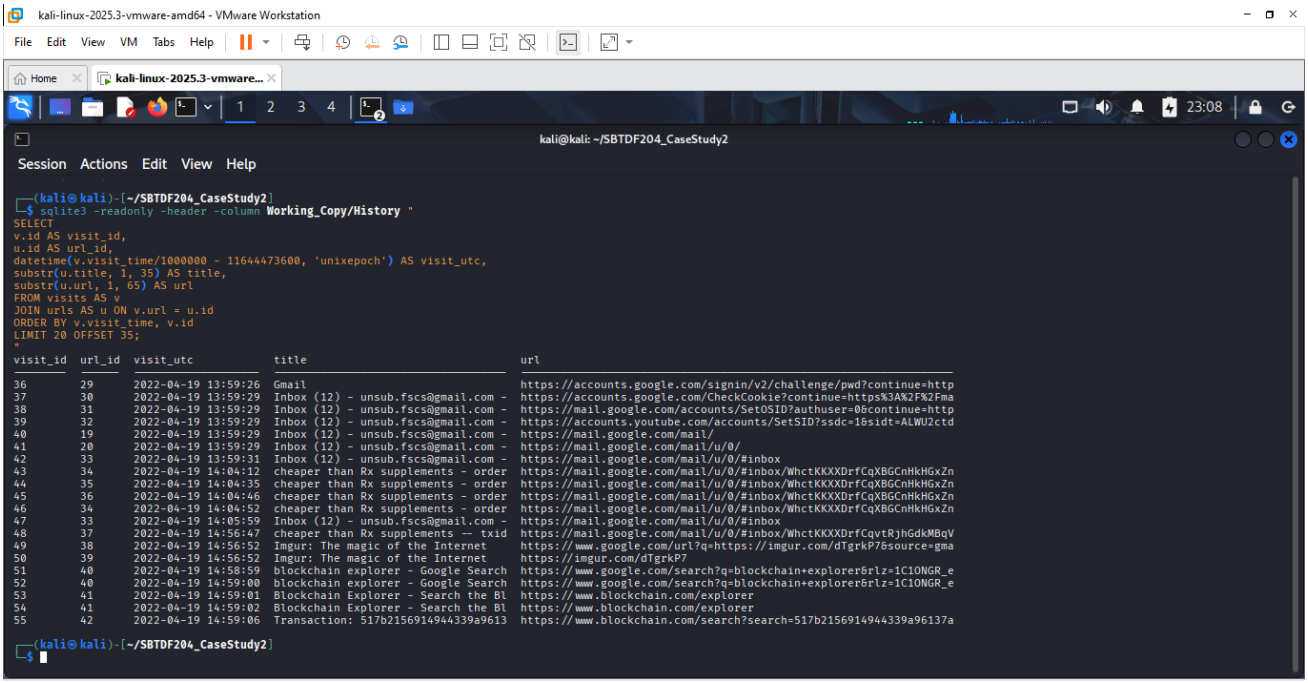


Fig 2.4 Chrome Browser History Records 36–55.

Evidence Log — E12

Field	Evidence Details
Evidence ID	E12
Source Record	Joined visits and urls tables; Visit IDs 36–55, URL IDs 19–42
Finding	Between 13:59:26 and 14:59:06 UTC on 19 April 2022, Chrome recorded Google account authentication pages, Gmail inbox activity, email pages titled “cheaper than Rx supplements - order,” access to Imgur, and searches for a blockchain explorer. The final records show navigation to Blockchain.com and a transaction-related page.
Why It Matters	These records establish a sequence of email-related and cryptocurrency-related browsing activity following the earlier Craigslist posting workflow. The email titles, image-hosting visits, and blockchain searches provide potentially relevant connections between communications and payment-related activity, although the records do not independently establish the contents of messages or completion of a transaction.
SQL or Screenshot Reference	Fig 2.4 — Chrome Browser History Records 36–55

Explanation

The screenshot shows further Chrome browsing activity involving Gmail account access, email pages with titles referring to "cheaper than Rx supplements - order," and subsequent navigation to Imgur. Later records contain searches for a blockchain explorer and visits to transaction-related pages, providing additional evidence for examining possible connections between communications and cryptocurrency activity.

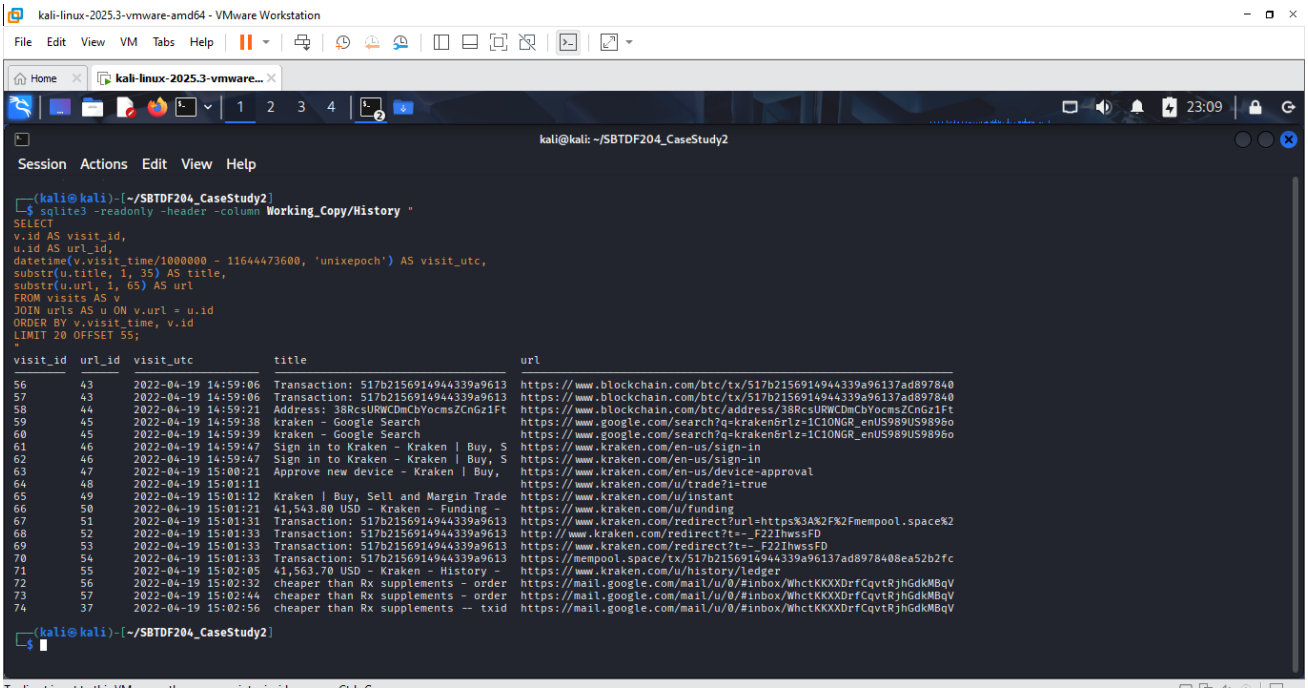


Fig 2.5 Chrome Browser History Records 56–74.

Evidence Log — E13

Field	Evidence Details
Evidence ID	E13
Source Record	Joined visits and urls tables; Visit IDs 56–74, URL IDs 37–57
Finding	Between 14:59:06 and 15:02:56 UTC on 19 April 2022, Chrome recorded visits to a Blockchain.com transaction page, a cryptocurrency address page, Google searches for Kraken, Kraken sign-in and device-approval pages, funding and trading pages, Mempool.space, and a Kraken transaction-history page. The final records show a return to Gmail pages titled “cheaper than Rx supplements - order.”
Why It Matters	The sequence establishes a connection in time between cryptocurrency-explorer activity, exchange-related browsing, and subsequent email activity. These records may support further investigation of the suspected payment workflow, but they do not establish wallet ownership, successful authentication, a completed cryptocurrency transfer, or the identity of the person operating the browser.
SQL or Screenshot Reference	Fig 2.5 — Chrome Browser History Records 56–74

Explanation

The screenshot displays the final 19 browser visits, including navigation through Blockchain.com transaction pages, Kraken account interfaces, and Mempool.space. The browser later returned to Gmail pages associated with the earlier subject lines, establishing the final recorded stage of the browsing sequence examined in this investigation.

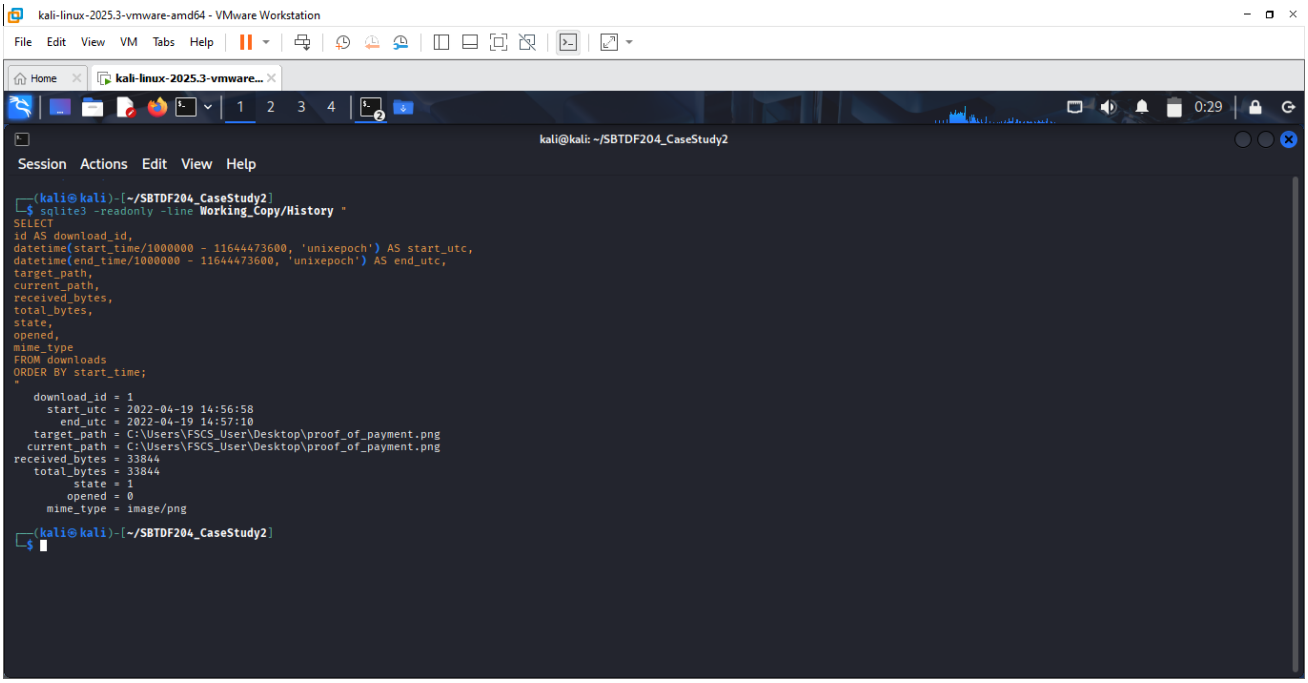


Fig 2.6 Chrome History Download Record Examination.

Evidence Log — E14

Field	Evidence Details
Evidence ID	E14
Source Record	downloads table; Download ID 1
Finding	The database recorded a PNG file named proof_of_payment.png, with a download start time of 14:56:58 UTC and end time of 14:57:10 UTC on 19 April 2022. The file was recorded as 33,844 bytes, with state = 1 (completed) and opened = 0.
Why It Matters	The filename suggests that the downloaded image may relate to payment documentation. The matching received and total byte counts, together with the completed status, support that Chrome recorded a completed download. However, the database does not establish the image's actual contents or whether a payment occurred.
SQL or Screenshot Reference	Fig 2.6 — Chrome History Download Record Examination

Explanation

The screenshot shows the examination of Download ID 1, which recorded an image named proof_of_payment.png between 14:56:58 and 14:57:10 UTC on 19 April 2022. The record showed a completed download of 33,844 bytes, although the image itself was not examined and its contents could not be independently verified.

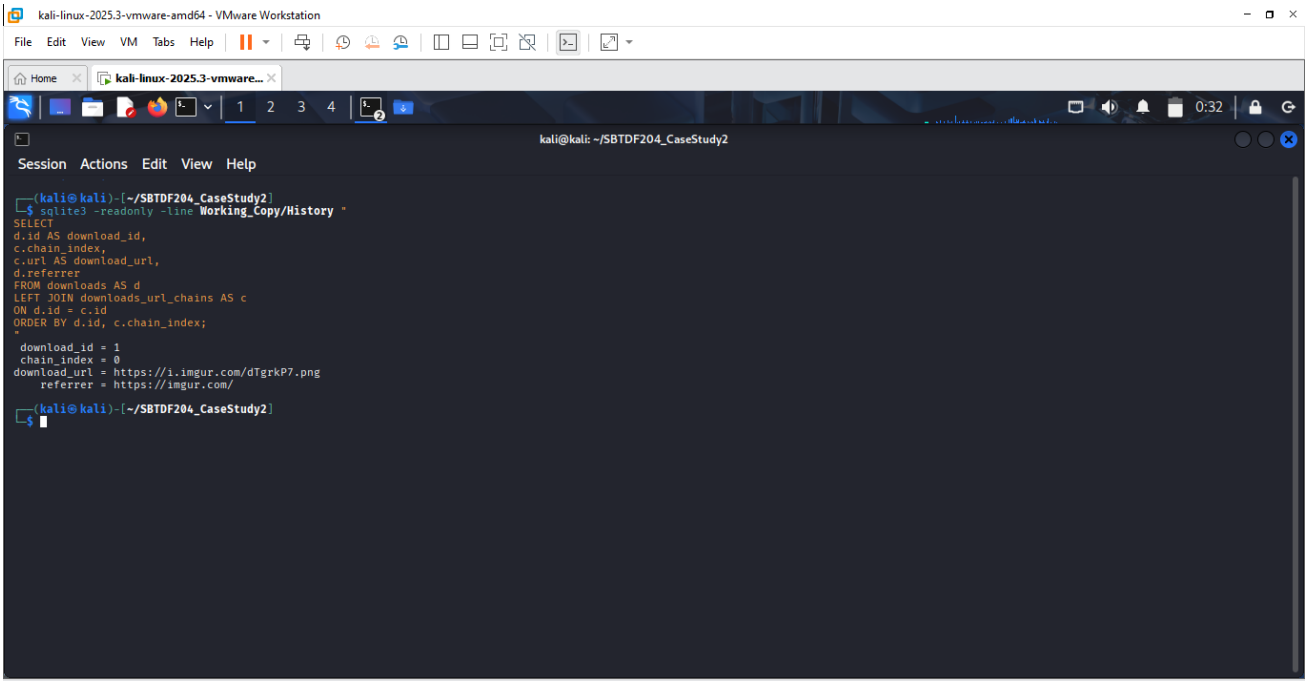


Fig 2.7 Chrome Download URL Chain and Source Identification.

Evidence Log — E15

Field	Evidence Details
Evidence ID	E15
Source Record	Joined downloads and downloads_url_chains tables; Download ID 1, Chain Index 0
Finding	The SQL query linked Download ID 1 to an image URL hosted on i.imgur.com. The recorded referrer was https://imgur.com/, and the URL-chain index was 0. Only one URL-chain record was identified for this download.
Why It Matters	The result establishes the recorded online source associated with proof_of_payment.png. When considered alongside the earlier Imgur browsing activity, it provides additional context for the download. However, the URL-chain record does not independently verify the image's contents or its relationship to an actual financial transaction.
SQL or Screenshot Reference	Fig 2.7 — Chrome Download URL Chain and Source Identification

Explanation

The screenshot presents the relationship between the recorded download and its associated URL-chain entry. The result identified an Imgur-hosted PNG address and an Imgur referrer, establishing the recorded download source without independently confirming what the image contained.

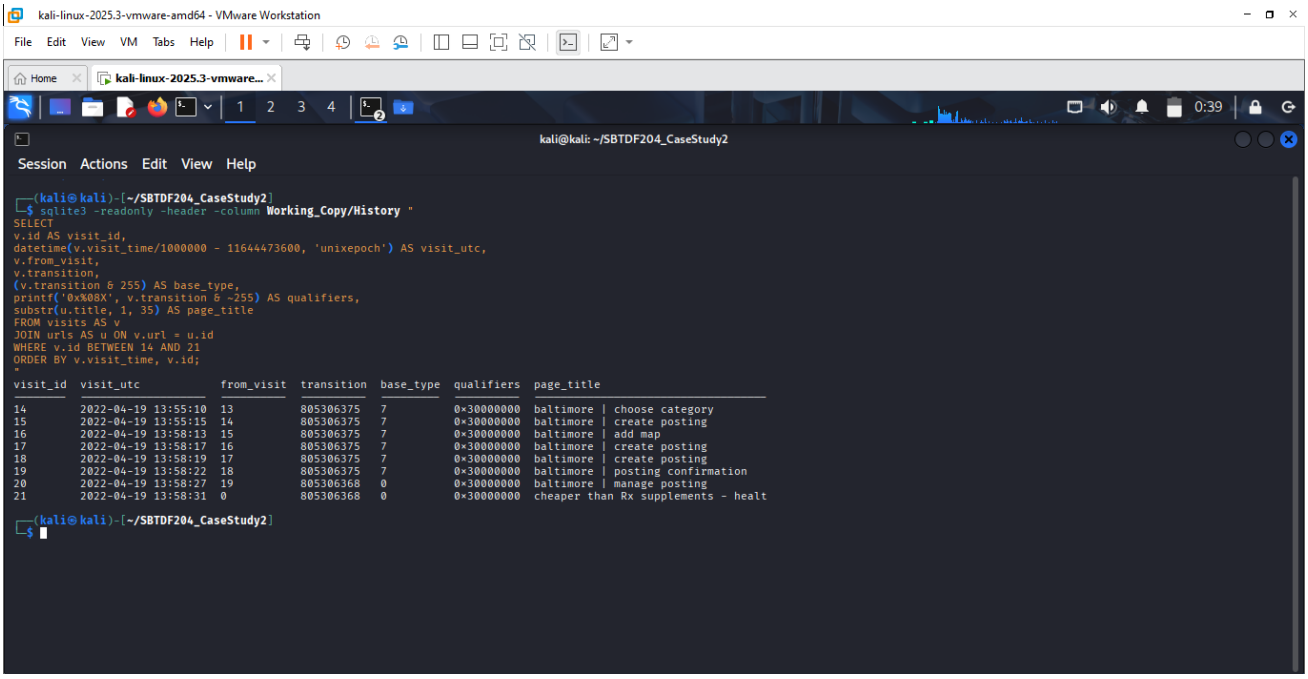


Fig 2.8 Craigslist Posting Navigation Chain and Transition Qualifiers.

Evidence Log — E16

Field	Evidence Details
Evidence ID	E16
Source Record	Joined visits and urls tables; Visit IDs 14–21
Finding	Between 13:55:10 and 13:58:22 UTC on 19 April 2022, Visit IDs 14–19 recorded transition value 805306375, corresponding to base type 7 (form submission). The records were connected through successive from_visit references. Visit ID 20 recorded a link transition associated with posting management, while Visit ID 21 recorded the listing titled “Cheaper than Rx Supplements.”
Why It Matters	The repeated form-submission transitions and linked visit records support a structured Craigslist posting workflow involving category selection, posting creation, map addition, and confirmation. Although the subsequent listing title is potentially relevant, the records do not establish the advertisement's complete contents or prove that controlled substances were offered for sale.
SQL or Screenshot Reference	Fig 2.8 — Craigslist Posting Navigation Chain and Transition Qualifiers

Explanation

The screenshot shows the navigation sequence associated with Craigslist posting activity, including repeated form-submission transitions and relationships between successive visits. These records connect category selection, posting creation, map addition, and confirmation, while the later listing titled "Cheaper than Rx Supplements" provides additional context for the suspected advertisement.

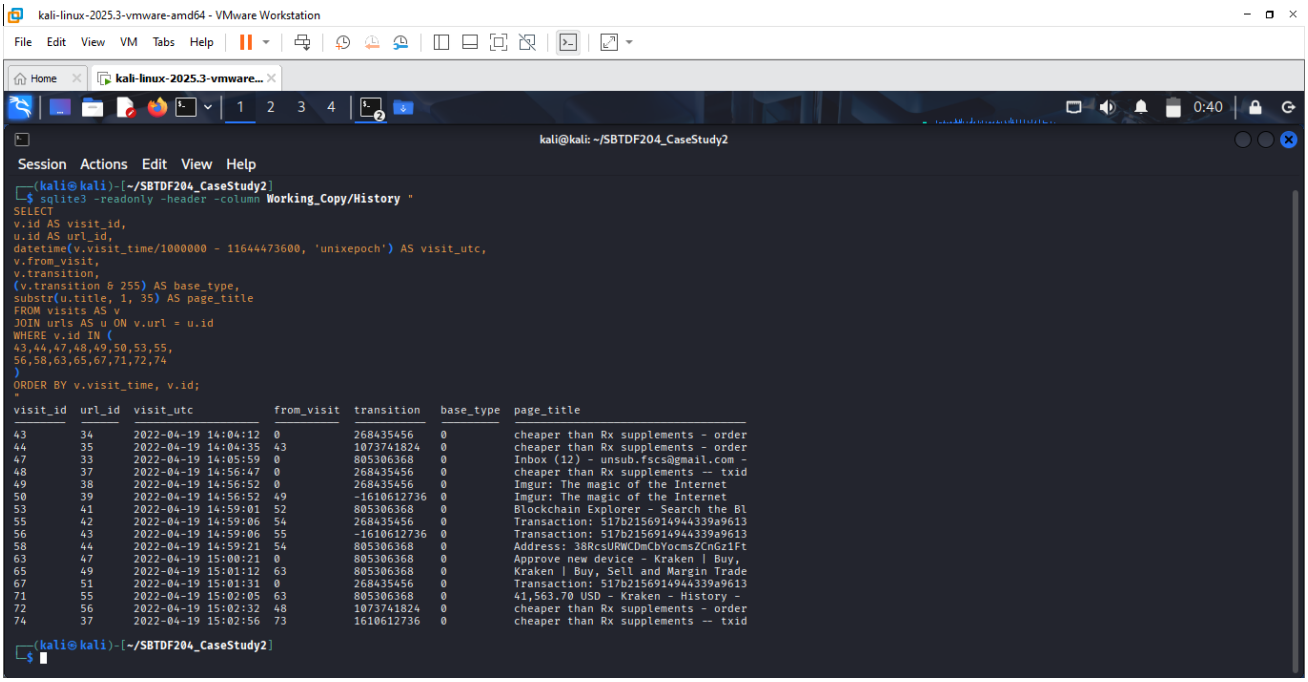


Fig 2.9 Communications and Cryptocurrency Navigation Analysis.

Evidence Log — E17

Field	Evidence Details
Evidence ID	E17
Source Record	Joined visits and urls tables; selected Visit IDs 43–74
Finding	The SQL query identified navigation records associated with Gmail, Imgur, Blockchain.com, Kraken, and transaction-related pages. All selected records had base transition type 0 (link), although their full transition values contained additional navigation qualifiers. The from_visit relationships identified connected events, including Visit IDs 43–44, 49–50, 55–56, and 63–65.
Why It Matters	The transition values and linked records help reconstruct how the browser navigated between communication-related, image-hosting, and cryptocurrency-related pages. Redirect qualifiers provide additional context for navigation behaviour, but these records alone cannot establish message contents, wallet ownership, payment completion, or the identity of the browser operator.
SQL or Screenshot Reference	Fig 2.9 — Communications and Cryptocurrency Navigation Analysis

Explanation

The screenshot presents selected navigation records associated with Gmail, Imgur, Blockchain.com, and Kraken, including timestamps, transition values, and preceding visit references. The results reveal several connected navigation events and redirect-related entries, helping explain how the browser moved between relevant webpages without establishing the completion of communications or financial transactions.

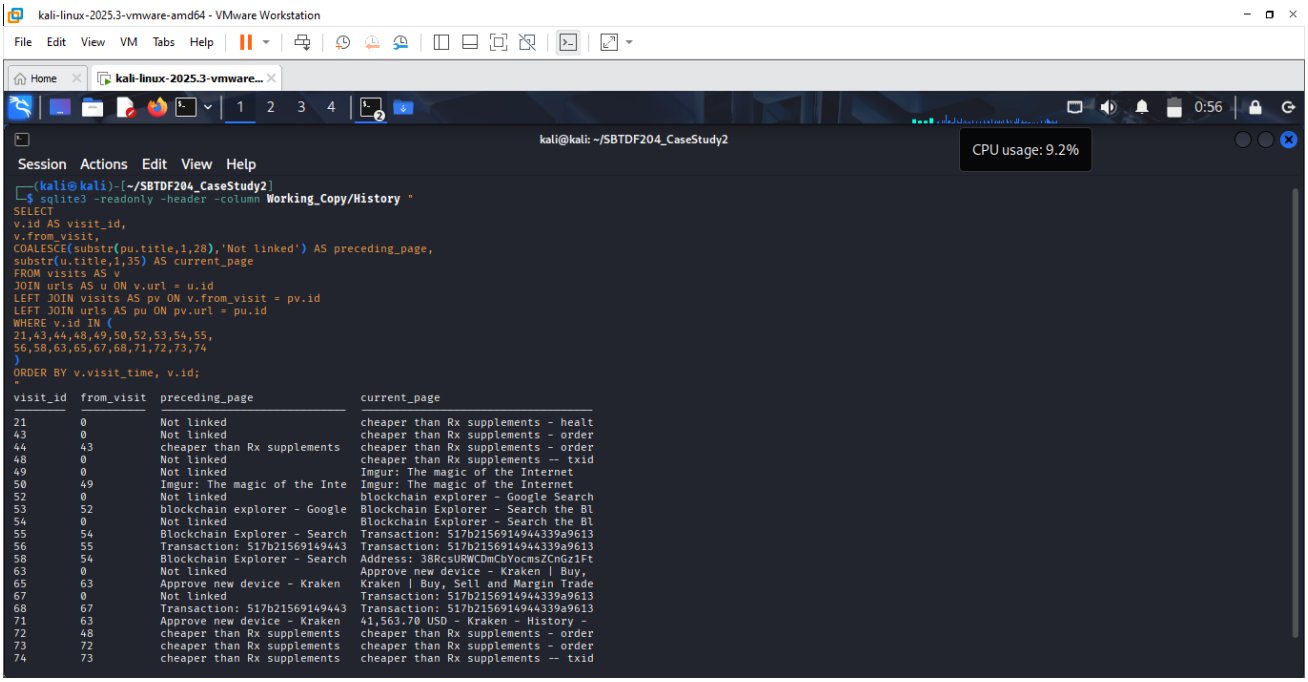


Fig 3.0 Chrome Browser Navigation Source and Destination Correlation.

Evidence Log — E18

Field	Evidence Details
Evidence ID	E18
Source Record	Joined visits and urls tables; selected Visit IDs 21–74
Finding	The SQL query identified navigation relationships between significant Craigslist, Gmail, Imgur, Blockchain.com, and Kraken records. These included Visit IDs 43–44, 49–50, 52–53, 54–56, 63–65, and 72–74. Several other records had from_visit = 0, indicating that no preceding visit was linked through that field.
Why It Matters	The results help reconstruct navigation sequences between relevant webpages. They support connections within recorded browsing sessions while distinguishing linked navigation from visits without an identified predecessor. This reduces the risk of incorrectly assuming that every chronological event was directly connected.
SQL or Screenshot Reference	Fig 3.0 — Chrome Browser Navigation Source and Destination Correlation

Explanation

The screenshot shows the results of an SQL query linking selected browser visits to their recorded preceding pages through the from_visit field. The findings establish relationships between several Gmail, Imgur, blockchain, and Kraken navigation events, while entries with no linked predecessor highlight the limitations of reconstructing every navigation step.

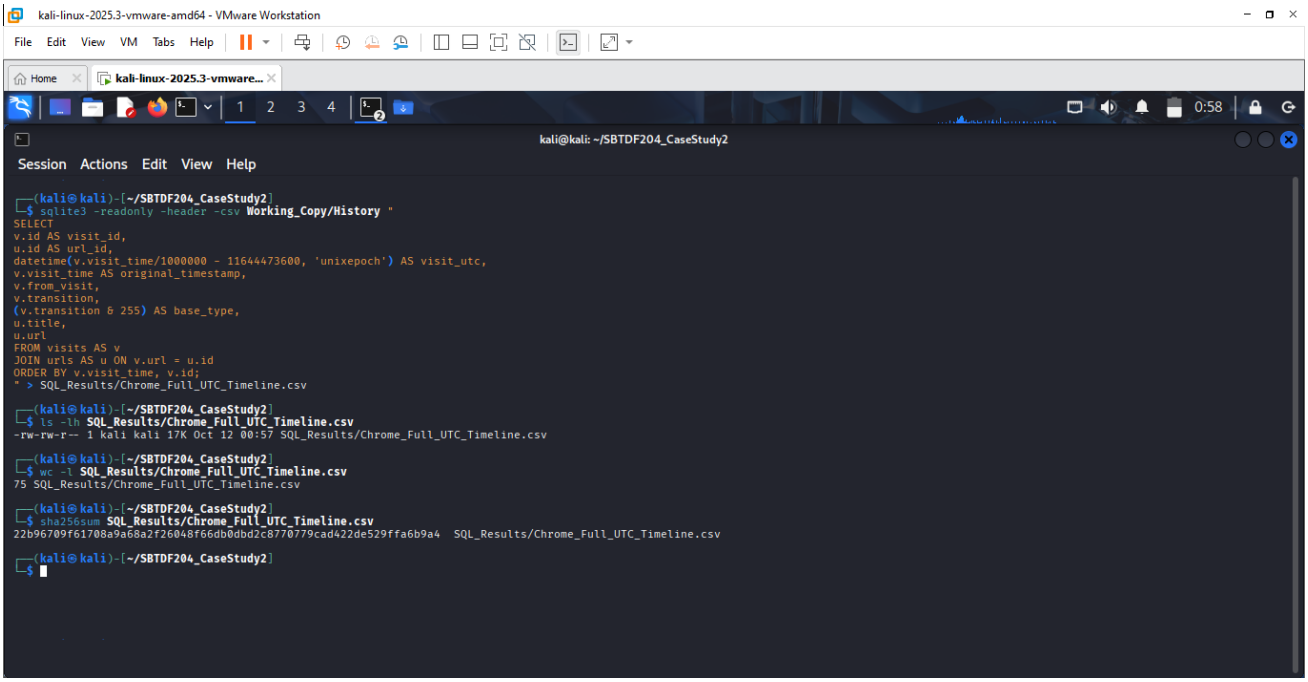


Fig 3.1 Complete Chrome Browser History Timeline Export and Verification.

Evidence Log — E19

Field	Evidence Details
Evidence ID	E19
Source Record	Working_Copy/History; exported file SQL_Results/Chrome_Full_UTC_Timeline.csv
Finding	The SQL query exported browser-history records into a CSV file containing visit IDs, URL IDs, original timestamps, converted UTC timestamps, navigation references, transition values, page titles, and URLs. The resulting file was approximately 17 KB and contained 75 lines, representing 74 visit records and one header row. A SHA-256 hash was also calculated for the exported file.
Why It Matters	The export preserves the extracted browsing information in a structured format suitable for chronological analysis, cross-referencing, and reporting. Its SHA-256 hash provides an integrity reference for the exported CSV, allowing later copies to be checked for changes.
SQL or Screenshot Reference	Fig 3.1 — Complete Chrome Browser History Timeline Export and Verification

Explanation

The screenshot documents the export of Chrome browser-history records into a CSV file containing visit identifiers, timestamps, transitions, page titles, and URLs. The resulting file contained 75 lines, representing 74 browser visits and one header row, and a SHA-256 hash was generated to provide an integrity reference for the exported information.

Limitations

Several limitations affected the interpretation of the recovered Chrome History evidence.

1. Limited evidence source

The investigation relied on the supplied Chrome History SQLite database. Other potential sources, such as recovered email content, complete webpage contents, account records, or the downloaded image itself, were not examined. This limited the extent to which the recorded browsing activity could be independently verified.

2. Browser activity does not prove human identity

Although the database records websites visited and related navigation events, it does not establish who was physically using the device. Shared devices, multiple users, or access to the same browser profile remain possible alternative explanations.

3. Incomplete communication evidence

The Gmail records contained page titles and URLs associated with email threads and compose-related navigation. However, they did not provide the complete messages or establish whether particular emails were successfully sent or received.

4. Unverified cryptocurrency transactions

The browser recorded transaction references and visits to cryptocurrency services, but the investigation did not independently verify the blockchain transactions or establish ownership of the associated wallets. The records therefore cannot confirm whether a payment was completed or received.

5. Downloaded image was not examined

The download record identified an image named proof_of_payment.png, but the actual file contents were unavailable for examination. Consequently, the filename could not be treated as confirmation that the image represented a genuine financial transaction.

6. Browser navigation may include redirects

Some recorded visits contained redirect qualifiers, while others lacked a linked preceding visit. These details show that not every history entry necessarily represents a separate deliberate action by the browser operator.

7. Possible gaps in the recorded history

The supplied database represents the browser-history information available for examination. Any activity not retained in the database, including deleted or otherwise unrecorded browsing, could not be reconstructed. The original device clock was also not independently verified.

Conclusion

The examination of the Chrome History database successfully reconstructed a sequence of browsing activities recorded on 19 April 2022. The investigation identified 74 browser visits involving Craigslist, Gmail, Imgur, Blockchain.com, Kraken, and Mempool.space, along with a completed download record for an image named proof_of_payment.png.

The recovered evidence shows a progression from Craigslist posting activity to email-related browsing, followed by image-hosting and cryptocurrency-related activity. Matching posting identifiers, transaction references, and image URLs provided additional support for relationships between several recorded events.

While these findings are consistent with aspects of the suspected drug trafficking scenario, they do not independently establish that controlled substances were sold, that cryptocurrency payment was completed, or that a particular individual carried out the activities.

Overall, the investigation achieved its objective of preserving and examining the supplied browser-history evidence, reconstructing the significant events, and identifying their possible relationships. Further independent evidence would be necessary to confirm the alleged criminal conduct or attribute the browsing activity to a specific person.

APPENDICES

The following appendices organise the supporting materials already produced during the investigation.

Appendix A — Forensic Screenshots
This appendix contains the 22 labelled screenshots, beginning with the preparation of the investigation workspace and ending with the export and verification of the complete browser-history timeline.The screenshots document the procedures performed in Kali Linux and provide visual support for the findings described in the report.

Appendix B — Evidence Log
The evidence log contains entries E01–E19, covering evidence preservation, database examination, browser-history extraction, navigation analysis, download investigation, and timeline export. Each entry identifies the source record, describes the finding, explains its relevance, and refers to the corresponding screenshot or SQL result.

Appendix C — SQL Examination Results
This appendix covers the SQL commands and results used to examine the database structure and extract relevant browsing records. The examination included table identification, schema inspection, record counting, UTC timestamp conversion, navigation transition analysis, download examination, and relationships between recorded visits.

Appendix D — Reconstructed UTC Timeline
The timeline appendix contains the chronological browser-history reconstruction derived from the 74 recorded visits. The supporting CSV file, Chrome_Full_UTC_Timeline.csv, contains the extracted visit identifiers, URL identifiers, timestamps, navigation information, page titles, and recorded URLs. The selected-event timeline and correlation workbook provide additional analysis of the significant records.

Appendix E — Evidence Integrity Verification
This appendix references the SHA-256 verification of the preserved Chrome History database and forensic working copy, together with the separately calculated hash for the exported timeline CSV. These records provide an integrity reference for the evidence examined and the derived timeline created during the investigation.

Screenshots

Fig 1.0 Forensic Investigation Directory Structure.
Fig 1.1 Original Chrome History Evidence Preservation and working copy.
Fig 1.2 Chrome History SHA-256 Integrity Verification.
Fig 1.3 Verify SQLite Installation.
Fig 1.4 Chrome History Database Identification and Integrity Check.
Fig 1.5 Chrome History SQLite Database Tables.
Fig 1.6 Chrome History URLs Table Structure.
Fig 1.7 Chrome History Visits Table Structure.
Fig 1.8 Chrome History Downloads Table Structure.
Fig 1.9 Chrome History Downloads URL Chains Table Structure.
Fig 2.0 Chrome History Database Record Counts.
Fig 2.1 Chrome Browser Visit History and UTC Timestamp Conversion.
Fig 2.2 Chrome Browser Navigation and Transition Analysis.
Fig 2.3 Chrome Browser History Records 16–35.
Fig 2.4 Chrome Browser History Records 36–55.
Fig 2.5 Chrome Browser History Records 56–74.
Fig 2.6 Chrome History Download Record Examination.
Fig 2.7 Chrome Download URL Chain and Source Identification.
Fig 2.8 Craigslist Posting Navigation Chain and Transition Qualifiers.
Fig 2.9 Communications and Cryptocurrency Navigation Analysis.
Fig 3.0 Chrome Browser Navigation Source and Destination Correlation.
Fig 3.1 Complete Chrome Browser History Timeline Export and Verification.

References

ICDFA Materials.